

closure_evidence_20260922

BOB-220 closure evidence — 2026-09-22

Fix

Root-caused: `/usr/bin/chmod` on this host is `utils-coreutils` (a Rust reimplementation), not GNU. Measured: `chmod 755 dir` AND `chmod 0755 dir` (both under 5 characters) LEAVE a directory's `setgid` bit set; only the fully-zero-padded 5-character `chmod 00755 dir` form (or explicit `g-s`) actually clears it. Files are unaffected either way (the kernel already clears `setuid/setgid` on `chown(2)` before this `chmod` runs).

`scripts/ownership_repair.sh (flush_batch()):` changed `_mode="$(printf '%o' "${_mode} ")"` to `_mode="$(printf '%05o' "${_mode} ")"`, with an in-source comment recording the measured `chmod` quirk. BOB-207's `uid-only-selection` fix does NOT make this moot — a foreign-`uid`-owned `setgid` directory is still selected for repair and still hits this code path.

RED/GREEN evidence (real end-to-end repair run, real foreign uid via `podman unshare chown 1:1`)

Pre-fix: mode stays 2755 after repair. Post-fix: mode becomes 0755.

Independently re-verified this session

```
$ bash tests/unit/test_ownership_repair_setgid_strip.sh
PASS: control: the pre-fix shape (bare '%o') reproduces the
```

reported no-op – 2755 survives under this identical harness,
proving Case 1 is discriminating
RESULT: 2 passed, 0 failed, 0 skipped

Plus the full tests/unit/test_ownership_repair.sh (146/146).

See docs/qa/B0B-208/closure_evidence_20260922.md for the combined
git diff --stat.